

Fourier transform activity

Student activity

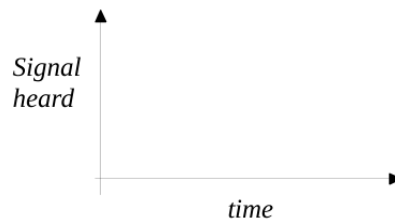
Name _____ Date _____

- Construct graphs of data using the ***FourierTransform_ActivityGraph.ods*** file.
- Each graph should have data that appears random but has data points at one or more regular time intervals.
- Time axis from 0 s to 60 s
- Each graph contains two signals at the periods indicated below, with noise included in the signal. These are created in the *FourierTransform_ActivityGraph.ods* file.

1. (3, 7)
2. (4, 7)
3. (3, 8)
4. (4, 9)
5. (4, 6)
6. (3, 5)
7. (5, 9)
8. (5, 7)
9. (6, 9)
10. (5, 8)

50 s on time-axis = 7.50" $\rightarrow$ 1 s = 0.15"

- Intro
- Hand out a blank graph of signal vs. time to each student.



- Demonstrate GNU Radio sound of single, clean periodic pulses at 1.0 s intervals. Ask students to sketch what they think the pulse would look like on the graph.
- Have students discuss their graph with another student; then discuss as class:
 - Have a volunteer sketch their graph on the board
 - Why does your graph represent what you heard?
 - How is the *period* of a wave represented in your graph?
 - This is an example of a periodic signal.

•Signals that are not noise usually have some sort of pattern that is repetitious - periodic.

•Review the *period*, *amplitude*, and *frequency* of this signal, and how each is represented in a signal.

- Then demonstrate GNU Radio pulses with noise added. Ask students to sketch what they think the signal would look like in this case.
- Students compare with a neighbor.
- Then discuss as class:

•What's different?

•What's the same?

•Noise is usually not periodic. Recall from yesterday: Signals carry information.

When processing signals, it is necessary to distinguish the signal from the noise.

•Recall Tad's presentation with the photo filtering. This is filtering of amplitude.

•Another method scientists use is filtering by frequency. Scientists have methods to filter signals by frequency (or period).

- Signal Processing Activity
- Demonstrate process with the teacher graph

•explain masks.

•demonstrate sliding process to identify periods.

•explain that a periodic signal must fill at each period translation.

•explain how noise shows up – signals less than 1.0 (usually), and not periodic.

•Explain offset.

- Group students into pairs.
- Handout signal sheets and masks. Explain that they are to determine the periods in each signal graph; then replot the data showing only those periodic signals, without noise.
- Have students sketch the Signal vs. Frequency graph that corresponds to their
- Post Activity Discussion
- What challenges are there determining the periodic signals in the total signal?
- For the 3 s and 4 s signals, harmonics show up. How do you think we can distinguish whether the actual signal is the 3 s, 6 s, or 9 s?
- Many signals are a mixing of multiple frequencies – next activity with Howard.
- Scavenger hunt clue.
- Motivation: Explain/remind students that waves are repetitious, e.g. water waves oscillate periodically. Information can be carried in the pattern, e.g. the morse code activity done previously.
- Show sample graphs:
- Example showing sum of sinusoidal signals. What periodic waves are in this?
- Example with periodic pulse signal in noise.

•What repetitious (periodic) waves in this signal?

•Demonstrate masking method for determining the periodic waves in this.

- Lay the mask over the graph and slide the mask until each opening aligns with the data at one second intervals.
- By sliding the mask to match up with the data at each one second interval the period of the signal can be determined.
- Pair students up. Hand out graphs and masks to each group.
- Students are to use the mask method to determine the period of any waves in their graph.
- Have each group sketch what the graph would look like if there were no noise.
- Extension: On a signal vs. frequency graph, have each group plot signal vs. frequency.
- Scavenger hunt:
 - column (0 – 9) determined by amplitude of stronger signal
 - row (X – R) determined by frequency of weaker signal; use a table to convert frequency to a letter.